

O'REILLY®

# Hands-on Prompt Engineering





# Poll

Who has used prompting and generative AI models for work tasks? ChatGPT, Gemini, Claude any other model?

- Use it all the time
- Use it sometimes
- Have never used it for work
- Have never user any generative AI chatbot

# Poll



What kind of tasks do you usually perform with generative AI?

- Email and content writing
- Programming and debugging
- Research into topics
- Personal use such as planning meals, trips, parties



# Prerequisites



- No prerequisites at all
- Course content geared towards a technical audience
  - Developers, testers, analysts, program and project managers - anyone who works in technology



# Prompt Engineering

Prompt engineering is the process of designing and refining prompts to effectively guide the output of language models and other AI systems.



# Importance of Prompting

- Guide the model to generate relevant output
- Improve quality and diversity of generated output
- Increase control and interpretability, reduces bias
- Mitigate **hallucination** by guiding the model
- Determine good and bad outcomes by goal setting





# Challenges in Prompting

- Get the required results on the first try
- Figuring out the right place to start
- Mitigating bias in the output
- Obtaining diversity and creativity in the results
- Maintain balance between precision and creativity





# Where Can We Use Prompt Engineering?

ChatGPT - AI chatbot that took the world by storm

Gemini - Google's conversational AI service

Claude - Anthropic's LLM and chatbot

Microsoft Copilot

Muse Spark - Meta's latest proprietary model

Grok - xAI's chatbot



O'REILLY®

# Understanding Generative AI and LLMs





# Generative AI



Generative AI refers to artificial intelligence systems that can create new content, such as text, images, music, or videos, by learning patterns from existing data.



# How Do Generative AI Models Work?

- Uses a powerful ML model to learn patterns and relationships in a dataset created by humans
- The model uses learned data to create new content
- The idea is that the new content resembles the content that the model has already seen before





# Large Language Models

Huge models

Large datasets

O'REILLY®

# What is GPT?





# ChatGPT and GPT

- ChatGPT is referred to as a model – but it is the interface to the underlying model that powers it
- ChatGPT is a chatbot built using a text-to-text model called GPT (**G**enerative **P**re-trained **T**ransformer)





# Generative Pre-trained Transformer

**Generative:** Refers to the model's ability to generate text

**Pre-trained:** Model pre-trained before fine-tuned for specific tasks

**Transformer:** Type of neural network architecture used by GPT



# GPT



Generative Pre-trained Transformers are a type of large language model (LLM) and a prominent framework for generative artificial intelligence.



# Other Models



Meta's **Llama** - another generative transformer-based foundational LLM

There are other foundational models beyond the GPT series - Google's **Gemini** or **PaLM**

Anthropic's **Claude**, another GPT-based model

O'REILLY®

# Crafting Prompts





# Anatomy of a Prompt





# Advanced Techniques in Prompting

Zero-shot prompting

Few-shot prompting

Chain-of-Thought  
prompting

Augmented knowledge  
prompting



# Three Steps in Prompt Engineering

Start with a  
reasonable prompt

Refine, iterate,  
evaluate, repeat

Calibrate and  
fine-tune





# Start with a Reasonable Prompt

Be precise and clear

Assign roles or personas

Use constraints

Avoid leading or biasing  
the model



# Refine, Iterate, Evaluate, Repeat

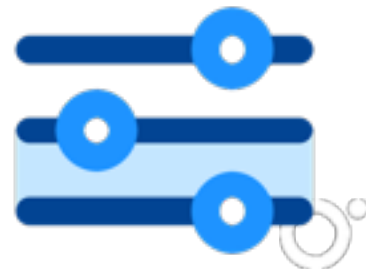
- Start somewhere with an initial draft
- Generate and test the response
- Evaluate if the prompt aligns with the objective
- Refine the prompt to guide model in the right direction





# Calibrate and Fine-tune

- Advanced techniques to better align the model for specific tasks
- Involves adjusting the model parameters to achieve this







# Best Practices for Prompt Design

- Make sure you are using the latest model e.g. ChatGPT uses GPT5
- Limited queries on the free version
- Put instructions at the beginning and separate instructions from text using ###
- Be specific, descriptive, and detailed about context, outcome, length, format, style
- Provide examples for what you want the output to look like





# Best Practices for Prompt Design

- Start with zero-shot and then use few-shot, and then fine-tune model
- Make descriptions crisp, clear, and unambiguous, and avoid vague language
- Specify "what to do" rather than "what not to do"

